

Spy Root Universal: A Client-Side Web Auditor for CERN ROOT Files

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Software

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Summary

In High-Energy Physics (HEP) collaborations, such as the ATLAS Experiment at CERN, the standard binary format for storing data and simulation outputs is the .root file format. Analyzing or simply inspecting these files traditionally requires a complete native installation of the ROOT framework, setting up virtual environments, or executing complex C++ macros or Python/Uproot scripts in a terminal.

Spy Root Universal is a lightweight, zero-installation, open-source web application designed to break this barrier. Operating 100% on the client-side within any modern web browser, it allows researchers to instantly map binary metadata structures, perform deep scans of TTree branches, and interactively plot embedded histograms directly from local storage without server-side processing or data transmission.

Statement of Need

During experimental data workflows and software quality accounting, researchers frequently need to audit data repositories, cross-check file integrity, and inspect metadata trees (such as checking multi-layered calorimeter containers). Existing tools require localized dependencies or command-line proficiency, which introduces latency for quick administrative data validations.

Spy Root Universal addresses this need by providing an agnostic and secure framework built entirely on web standards. By directly interfacing with the binary internal key arrays (fBranches.arr) via the JSROOT ES6 library, the application uncovers complex, deeply nested xAOD structures that typical surface-level web tools fail to map. Since the execution happens entirely within the browser's local sandbox memory, it ensures absolute privacy and data protection, fulfilling data governance criteria for scientific data packages.

State of the Field

Traditional inspection of ROOT files relies primarily on the native C++ ROOT environment or heavy Python stacks using uproot and awkward. While tools like JSROOT provide excellent standard web rendering, they are typically implemented as components within larger, server-dependent monitoring systems. Spy Root Universal fills a distinct niche by wrapping these capabilities into a standalone, zero-config auditing tool optimized specifically for client-side multi-file validation and immediate structural reporting without any server backend.

35 Software Design

36 The software is implemented using modern web standards (HTML5/ES6+) and leverages the
37 core decompression and deserialization capabilities of JSROOT. The architecture is strictly
38 decoupled into a UI layer managed by responsive components and a background processing
39 controller that handles the hierarchical traversal of TDirectory trees and TTree metadata
40 blocks. Explicit memory management is performed via localized lifecycle hooks and `cleanup()`
41 calls to ensure browser stability when mapping large files.

42 Research Impact Statement

43 By eliminating the technical friction of framework installation, Spy Root Universal accelerates
44 the data verification pipeline for researchers working within large HEP collaborations. It enables
45 quick quality control checks of simulation outputs—such as Minimum Bias event files or
46 calorimeter occupancy data—directly from any device with a web browser. This drastically
47 lowers the entry barrier for technical audits, data governance compliance, and cross-laboratory
48 file validations.

49 AI Usage Disclosure

50 The authors confirm that no generative AI or automated writing assistants were utilized in the
51 conceptualization, development, or drafting of the software and manuscript.

52 Mentions

53 This tool builds upon the infrastructure provided by the CERN JSROOT library (Linev & Belov,
54 2024), adapting it for robust, recursive directory scanning (TDirectory), automated memory
55 management (`cleanup()`), and streamlined structural text-report generation for repository
56 auditing.

57 References

58 Linev, S., & Belov, B. (2024). *JSROOT: JavaScript ROOT v7.x*. CERN. [https://doi.org/10.](https://doi.org/10.5281/zenodo.1326402)
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